

# HARI KRISHNA

Software Engineer | AI/ML

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## SUMMARY

AI/ML and Full-Stack Software Engineer with hands-on experience building production-oriented systems — from computer vision and deep learning pipelines to LLM integration and real-time web applications. Comfortable owning a project end-to-end: model training, REST/WebSocket API design, and deployed, user-facing interfaces, using Python, PyTorch, TensorFlow, React, FastAPI, Flask, and Django. Top-8% global finisher (#2632 of 33,953) in ICPC 2026 Online Challenge 1 powered by Huawei.

## SKILLS

**Languages:** Python, Java, JavaScript, SQL, C++

**AI/ML:** Machine Learning, Deep Learning, Computer Vision, NLP, PyTorch, TensorFlow, Scikit-learn, U-Net, YOLO

**AI/ML Tools:** Transformers, llama.cpp, Faster Whisper, DeepFace, MiDaS

**Frameworks:** FastAPI, Flask, Django, Django REST Framework, React, Node.js

**Backend & APIs:** REST APIs, WebSockets, Socket.IO, TCP/JSON, Authentication, JWT

**Libraries:** OpenCV, NumPy, Pandas, Rasterio, GDAL

**Databases:** MongoDB, MySQL

**Tools:** Git, GitHub, Linux, Docker

**Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

## EXPERIENCE

**Cybersecurity Intern — CDAC, Noida** *Aug 2024 – Oct 2024*

- Analyzed suspicious files in an isolated virtual-machine environment using Cuckoo Sandbox, examining runtime behavior, system calls, processes, and network activity.
- Correlated sandbox findings with VirusTotal results to investigate and profile malware samples.

**Cybersecurity Trainer — LBSCEK, Kasaragod** *Mar 2025*

- Delivered hands-on cybersecurity training to 50+ students covering cryptography from fundamentals through advanced concepts.
- Led practical exercises involving encryption, Python scripting, and cybersecurity concepts.

**Full-Stack Developer Intern — Hexinox, Kochi** *Jul 2025*

- Developed full-stack web applications using React, JavaScript, MongoDB, and Supabase, including a database-driven shopping platform with user-facing forms and authentication workflows.
- Implemented CRUD workflows, REST API integrations, authentication, login, and password-protected account functionality, including a forgot-password flow.
- Integrated Supabase certificate upload and storage workflows with MongoDB-backed application data.
- Deployed web applications through Netlify using Git/GitHub and validated frontend/backend functionality through browser and network inspection.

## PROJECTS

**SlipSense — Landslide Susceptibility & Runout Prediction System** — *Python, Machine Learning, U-Net, FastAPI, React, GIS*

[GitHub](#)

- Engineered a pixel-level landslide susceptibility pipeline for Kerala using 9 DEM-derived terrain features, including slope, aspect, flow accumulation, TWI, SPI, relief, drainage density, and river proximity.
- Implemented a two-stage ML/DL workflow combining Random Forest and U-Net to generate susceptibility rasters and spatially refine hazard predictions, plus D8 flow-based runout modeling to trace debris transit and deposition zones.
- Built a FastAPI/rio-tiler tile-serving backend and a React GIS dashboard with LeafletJS 2D mapping and a CesiumJS 3D terrain viewer, supporting pixel inspection, weather-based risk scoring, and district-level alerts.
- Implemented an SMS alert system (Twilio/Fast2SMS) that automatically notifies districts when susceptibility, rainfall, and hazard-zone thresholds are jointly exceeded.
- Exposed 9 REST endpoints — tile serving, pixel inspection, weather proxy, and district-level alert management — and built an interactive frontend with layer toggles, opacity sliders, and a per-district filter for historical susceptibility data.

- Classified terrain into a 4-tier hazard system (Safe, Deposition, Transit, Failure) with custom raster colorization, computing a combined risk score from susceptibility and real-time rainfall.
- Sourced and processed CartoDEM/SRTM elevation data, GSI/KSDMA historical landslide records, and live OpenWeather API data to train, validate, and drive the pipeline.

#### **SecureVision AI — Surveillance Platform** — *Python, Flask, React, MongoDB, YOLO*

[GitHub](#) | [Live Demo](#)

- Architected a real-time surveillance pipeline integrating 5 deep-learning models — YOLOv8 Pose, MiDaS, SlowFast R50, DeepFace, and RetinaFace — for activity recognition, pose estimation, depth-based height estimation, and face identification.
- Designed a Flask/React monitoring architecture using REST APIs, Socket.IO/WebSockets, and MJPEG streaming to deliver annotated video, real-time events, metrics, and persistent security-event data.
- Built the dashboard with React 18, Vite, Recharts, and Framer Motion on a Flask-SocketIO + Waitress backend, replacing legacy camera monitoring with automated risk scoring based on posture and activity.
- Exposed REST endpoints for historical metrics and event review alongside the live WebSocket feed, so operators can monitor in real time and audit past incidents from the same dashboard.

#### **AI Talking Tom — Local AI Virtual Companion** — *Python, Godot, Llama.cpp, Qwen 2.5, Faster Whisper, Piper TTS, DeepFace, MongoDB, React*

[GitHub](#)

- Engineered a local AI companion across 20+ Python service modules, combining fully offline LLM inference, speech recognition, text-to-speech, emotion analysis, and conversational memory.
- Integrated Qwen 2.5 3B via Llama.cpp with Faster Whisper, Piper TTS, DeepFace, YOLOv8, and wav2vec2-based emotion analysis, fusing face and voice signals for multimodal interaction.
- Built a memory system that retains user facts and preferences across sessions and a dynamic personality layer that evolves with conversation history.
- Connected the Python AI backend to a Godot 4.6 avatar with 62 animations through TCP/JSON and built a React/Vite monitoring dashboard exposing Tom's internal state.
- Packaged the system to run fully offline on consumer hardware, with speech, vision, and emotion models fetched and cached automatically on first run.

#### **Voice HR Bot — AI-Powered Interview Simulator** — *Python, Django, Django REST Framework, React, Gemini 2.5 Flash, Web Speech API*

[GitHub](#)

- Developed an AI interview simulator combining Gemini 2.5 Flash with role-aware prompting, conversational context, and browser-based voice interaction.
- Implemented a React/Vite frontend with a Django REST backend and /api/chat/ endpoint for real-time communication between the application and AI service.
- Integrated speech recognition, speech synthesis, CORS/environment configuration, and production deployment using Netlify and Render/Gunicorn.

## **CERTIFICATIONS**

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- Machine Learning Specialization — Stanford Online / DeepLearning.AI
- TensorFlow Developer Professional Certificate — DeepLearning.AI
- Foundations of Machine Learning — NPTEL
- Ethical Hacking & Penetration Testing — C-DAC
- Cisco Networking Basics

## **ACHIEVEMENTS**

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- Ranked #2632 of 33,953 — ICPC 2026 Online Challenge 1 powered by Huawei (Codeforces), scoring 13407.172 points on an edge-cloud LLM inference scheduling problem
- 2nd Place — MediHack 2024 Hackathon
- NASA Space Apps Challenge — Participant

## **EDUCATION**

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B.Tech in Computer Science & Engineering, LBS College of Engineering, Kasaragod — CGPA: 7.42/10 · 2026 Graduate